

# SMART FLOOD- ALERT BRIDGE

Protecting Our Community, One Drop at a Time! 🌊

🏠[Your School Name]

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ADD PROJECT TEAM PHOTO OR MODEL  
PHOTO

# OUR JOURNEY TODAY



**1. The Crisis: Why floods hurt us**



**2. Our Idea: The Smart Bridge**



**3. The Tech: Sensors & Alarms**



**4. The Safety: Protecting Everyone**



**5. The Future: Scaling Up!**

# WHEN THE RIVER RISES...



During heavy monsoon rains, our local river overflows quickly and floods the bridge.



The bridge goes completely underwater, making it impossible for anyone to cross safely.



Result: Our entire village gets **cut off** from the rest of the world for hours or even days!



ADD PHOTO OF FLOODED BRIDGE OR LOCAL  
AREA

# THE HUMAN IMPACT



## Students

Miss school days, important exams, and classes when the bridge is flooded.



## Emergency Services

Ambulances and rescue teams face life-threatening delays reaching patients.



## Commuters

Vehicles and daily workers are stuck for hours with no alternate path home.



## Local Residents

Families are cut off from markets, hospitals, and workplaces for days.

# OUR SOLUTION: THE SMART BRIDGE



A bridge that "thinks" and "acts" before the flood hits.



It senses water levels and reacts automatically to keep the path safe.



No more guessing! No more getting stuck in the flood!



ADD DIAGRAM OR 3D MODEL  
OF THE SMART BRIDGE

# THE 3-STEP SAFETY FLOW



## 1. SENSE

The water sensor detects the rising river level.



## 2. ALERT

The buzzer beeps and LEDs glow to warn everyone.



## 3. PROTECT

The bridge deck lifts up to stay safe above the water!

# WHAT'S POWERING THE BRIDGE?



ARDUINO UNO

## Arduino UNO

The **"Brain"** that reads data and controls the bridge.



WATER SENSOR

## Water Sensor

The **"Eyes"** that watch the river level constantly.



BUZZER & LED

## Buzzer & LED

The **"Voice"** that warns people with sound and light.



POWER SOURCE

## Power Source

The **"Energy"** that keeps the whole system running.



MODEL PHOTO: FRONT VIEW



CIRCUIT DIAGRAM / CLOSE-UP

## SEE IT IN ACTION!



- ✓ Built with **affordable materials** like cardboard, wood, and simple electronics.
- ✓ Features **real-time sensing** that reacts instantly to rising water levels.
- ✓ A **scalable design** that can be adapted for real-world bridges in our community.



# BENEFITS FOR EVERYONE



## Early Warning

Alerts people before the bridge becomes dangerous, giving them time to stay safe.



## Prevents Accidents

Reduces the risk of vehicles or pedestrians accidentally entering deep flood water.



## Saves Time

People can plan alternate routes early instead of waiting blindly at a flooded bridge.



## Low Cost

Built using affordable components, making it easy for any village to install and use.

# WHAT'S NEXT?



## Mobile Alerts

Send real-time SMS warnings directly to villagers' phones before the flood hits.



## Solar Power

Use energy from the sun to keep our bridge sensors running 24/7, even during power cuts.



## IoT Network

Connect all bridges to a central hub so authorities can monitor every river at once.



**ANY QUESTIONS?** 🤔

**We'd love to hear your thoughts!**

# A SAFER TOMORROW



*"Small sensors, big safety. Let's build a flood-free future where no one is left behind!"*



ADD TEAM PHOTO OR THANK YOU IMAGE



Anjum Shaikh



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